

MANGILSAN, Korea, Republic of

WMO: 471260

Lat: 36.937N

Lon: 126.444E

Elev: 200

StdP: 98.95

Time Zone: 9.00 (E09)

Period: 86-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.3	(c) -8.0	(d) -15.6	(e) 1.0	(f) -7.9	(g) -14.0	(h) 1.1	(i) -6.7	(j) 12.7	(k) -2.5	(l) 11.2	(m) -1.8	(n) 5.1	(o) 340	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.8	(c) 30.2	(d) 25.0	(e) 29.1	(f) 24.4	(g) 28.0	(h) 23.7	(i) 25.8	(j) 29.0	(k) 25.2	(l) 28.1	(m) 24.5	(n) 27.1	(o) 3.7	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.5	27.9	24.1	19.5	26.9	23.8	19.1	26.4	81.2	29.3	78.1	27.7	75.2	27.8	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.1	(b) 10.9	(c) 9.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 13.1	(b) 10.9	(c) 9.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	-1.4	0.4	4.4	10.3	16.0	20.4	24.0	25.0	21.0	15.3	8.4	1.4
	DBStd	9.64	3.62	3.73	3.01	3.58	2.83	2.44	2.36	2.16	2.47	2.97	4.10	4.22
	HDD10.0	1185	354	270	175	38	1	0	0	0	0	3	77	268
	HDD18.3	2807	612	503	433	241	82	9	0	6	98	297	526	
	CDD10.0	1973	0	0	1	48	187	311	433	466	329	168	29	1
	CDD18.3	554	0	0	0	1	11	69	175	208	86	5	0	0
	CDH23.3	2836	0	0	0	5	26	217	943	1390	249	6	0	0
	CDH26.7	581	0	0	0	1	1	29	197	328	24	0	0	0
Wind	WSAvg	4.2	4.1	4.5	4.9	4.8	4.2	3.5	3.9	3.6	3.8	4.2	4.7	4.6
Precipitation	PrecAvg	1279	24	32	46	79	96	155	283	278	148	53	51	34
	PrecMax	1794	66	85	124	191	214	362	663	907	399	180	161	64
	PrecMin	808	5	3	9	23	17	29	111	29	12	2	10	6
	PrecStd	267	15	22	28	40	46	88	119	195	112	47	36	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	13.2	15.2	22.9	25.9	29.1	32.8	32.5	29.0	24.0	19.0	13.0
		MCWB	6.6	9.1	10.1	15.6	17.8	21.1	25.5	25.4	23.5	17.3	14.9	9.3
	2%	DB	6.1	9.1	12.4	19.8	23.8	27.2	30.1	30.9	27.1	22.1	17.0	10.1
		MCWB	3.9	6.4	8.2	13.6	16.6	20.4	25.1	25.4	22.0	16.5	13.5	7.3
	5%	DB	5.0	7.0	10.9	17.9	22.0	25.9	28.8	29.8	25.9	21.0	15.7	8.2
		MCWB	3.0	4.8	7.3	12.4	15.9	20.1	24.6	25.0	21.3	16.2	12.3	6.1
	10%	DB	4.0	5.8	9.1	15.9	20.2	24.2	27.1	28.5	24.9	19.9	14.0	7.0
		MCWB	2.2	3.8	6.0	11.0	15.2	19.2	23.7	24.4	20.7	15.1	11.2	4.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	10.2	11.0	16.7	19.6	23.2	26.7	26.8	24.8	19.3	16.2	10.1
		MCDB	9.5	10.9	14.2	21.2	23.2	25.9	30.3	30.1	27.4	22.3	18.2	12.8
	2%	WB	4.2	6.7	9.0	14.6	18.1	22.0	25.7	26.0	23.5	17.7	14.2	7.9
		MCDB	5.5	8.0	12.2	18.2	21.2	24.7	28.9	29.4	25.8	20.8	15.7	9.6
	5%	WB	3.1	5.1	7.4	13.0	17.0	21.1	25.0	25.4	22.4	16.8	12.8	6.3
		MCDB	4.6	6.5	10.2	16.7	20.6	24.0	27.8	28.5	24.7	19.8	15.0	8.1
	10%	WB	2.1	3.6	6.2	11.7	16.0	20.2	24.3	24.9	21.3	15.8	11.3	4.8
		MCDB	3.8	5.6	8.6	15.2	19.5	23.0	26.6	27.8	23.8	18.7	13.7	6.6
Mean Daily Temperature Range	5% DB	MDBR	6.7	6.2	7.0	8.1	7.5	6.8	5.2	5.8	6.2	7.1	6.5	6.0
		MCDBR	7.5	8.2	9.8	10.5	9.5	8.8	7.0	7.1	7.3	8.3	7.6	7.6
		MCWBR	6.5	6.4	7.3	6.7	6.0	5.5	4.0	3.8	4.2	5.1	5.4	6.5
	5% WB	MCDBR	6.8	7.5	9.3	9.7	8.6	7.6	5.9	6.4	6.3	7.2	6.8	7.4
		MCWBR	6.4	6.5	7.3	6.7	6.1	5.0	3.8	3.9	4.3	5.1	5.5	6.6
Clear-Sky Solar Irradiance	taub	0.429	0.473	0.547	0.580	0.599	0.668	0.633	0.558	0.493	0.482	0.465	0.429	
	taud	1.975	1.897	1.758	1.732	1.749	1.672	1.803	1.991	2.094	1.984	1.970	1.988	
	Ebn at Noon	711	732	716	723	719	669	689	734	761	716	668	677	
	Edh at Noon	137	167	210	227	227	245	214	173	148	151	137	127	
All-Sky Solar Radiation	RadAvg	2.43	3.26	4.39	5.10	5.66	5.25	4.30	4.55	4.24	3.66	2.45	2.11	
	RadStd	0.16	0.28	0.42	0.36	0.55	0.57	0.66	0.66	0.44	0.27	0.26	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+91	+53

Nomenclature: See separate page